Finite-Shot Quantum Reservoir Computing on Lorenz-96 Against Tuned Classical Baselines

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Abstract

Does a quantum reservoir computer (QRC) forecast a chaotic system better than a classical model when the comparison is fair? We fix four conditions: every classical baseline is tuned with a search budget comparable to the QRC's, the classical models see the same input window and have the same number of features, the QRC's headline number is read from a finite number of measurement shots, and every result is a multi-seed statistic. Under these conditions we run eight cells on one-step forecasting of a six-variable Lorenz-96 system with a twelve-qubit reservoir: two memory schemes (reset and recurrent), two Hamiltonians (a disordered XXZ model and an all-to-all Ising model) and two readouts (one measurement basis or two). In all eight cells the QRC loses to a random-feature map of the same width and window on every one of five fresh data seeds. The best QRC cell ties a degree-2 polynomial of the input window, and an echo-state network of the same width wins every recurrent cell by a wide margin. A second, non-commuting measurement basis is the QRC's largest gain but does not close the gap. In one single-basis XXZ pair per memory scheme, finite shots cost the recurrent QRC far more than the reset one and turn a tie with the random features into a loss. Neither a local refinement stage after the Bayesian search (tested in the best QRC cell) nor widened parameter ranges (tested in the two-basis Ising cells) changes a verdict against the random features or the echo-state network. In free-running forecasts no model's mean valid prediction time reaches one Lyapunov time, and the QRC stays behind the random features and the echo-state network. All tables are generated from the stored result records.

1 Introduction

Quantum reservoir computing trains only a linear readout on the measured outputs of a fixed quantum system driven by the input signal [1]. Because nothing quantum is trained, it is a candidate for near-term hardware, and several studies report that a small quantum reservoir forecasts chaotic or nonlinear time series better than a classical one [2, 3, 4]. Whether such a result says something about the quantum system or about the comparison depends on how the classical side was set up. We call a comparison fair when four conditions hold:

1. every classical baseline is tuned with a budget comparable to the QRC's, by the same search method, on the same tuning seeds;
2. the classical models see the same input window as the QRC, and those built to match it have the same number of features;
3. the QRC's headline number is computed from a finite number of measurement shots, as hardware would report it, not from exact expectation values;
4. every reported number is a statistic over fresh data seeds and reservoir seeds that were never used in tuning, with paired tests corrected for multiple comparisons.

What is new. In earlier work [5] we compared QRCs with capacity- and tuning-matched classical models in noiseless statevector simulation at up to eleven qubits, on coupled Hénon maps and a regime-switching autoregression, and found that neither a correlator readout nor measurement feedback gave the QRC an edge. A companion study tested whether a high-dimensional quantum reservoir gains from its dimension [6], and a third compared fixed-reservoir and variational quantum models on Lorenz-63 [7]. This paper changes five things. Every QRC number is read from finite shots, and the cost of shots is measured against exact values. A second, non-commuting measurement basis is added at the same total shot budget, and the width-matched classical models grow with it. The tuning budget is matched by search density (trials per tuned knob, run to convergence) rather than by trial count, and its sufficiency is checked by a local refinement stage and by widened ranges. Our own records show a verdict reversing when one ESN knob is added to the search. And every frozen model is also run as a free-running forecaster.

Finding. On one-step forecasting of Lorenz-96 with $K = 6$ variables and a $q = 12$ qubit reservoir, the QRC loses to a random tanh feature map of the same width and window on every data seed of all eight cells. Its best cell ties a degree-2 polynomial of the input window, and an echo-state network (ESN) of the same width wins every recurrent cell. The free-running test keeps this ordering.

2 How advantage claims are usually made

Six patterns recur in QRC benchmarks. None makes a result wrong; each makes it answer a narrower question than an advantage claim needs. We map papers to patterns only where their own text states the relevant fact.

1. Resources counted in different units. A quantum reservoir is often sized in qubits and a classical one in neurons. Ahmed, Tennie and Magri [2] find that on Lorenz-63 their best quantum architecture at 9 to 11 qubits reaches valid prediction times similar to those of classical reservoirs of 500 to 2000 neurons, while the classical reservoir outperforms every quantum architecture at the same reservoir size; three quantum architectures outperform it on a ten-variable Lorenz-96 system as the size grows (their Figs. 8 and 12). What a linear readout sees is the number of measured features, and we match that number directly.

2. Untuned or lightly tuned baselines. The strength of a baseline depends on which knobs are searched. Kobayashi, Fujii and Yamamoto [3] report the best ESN over a grid of five spectral radii and four learning rates (their Appendix D). Nguyen and Soh [4], forecasting the Mackey-Glass series with a driven Jaynes-Cummings reservoir, tune an ESN's spectral radius, leak, input scale and bias over 810 settings on the training loss, while their quantum model receives grid search, Bayesian search and a further QFI-aware optimisation stage.

3. QRC compared with QRC. Careful work compares quantum reservoirs across dynamical regimes [8] or measurement protocols [9]. That is how a reservoir should be designed, but a gain over another QRC is not a gain over a classical model.

4. Capacity metrics in place of forecast error. Hu et al. [10] give a framework for the expressive capacity of a quantum system under the sampling noise of finite measurement. Capacity guides design; it does not say whether a tuned classical model forecasts the task better.

5. Exact expectations in the headline, shots elsewhere. Hardware returns counts. The results of [2] come from noise-free emulation, with finite sampling left to future work; [3] state their main results under negligible statistical fluctuations and treat a finite number of measurements in their Appendix C; [4] obtain their margins from expectation-value simulations and analyse shots in a supplementary note.

6. Few seeds. A single reservoir draw on a single series gives no estimate of the spread between draws. We do not attribute this pattern to specific papers; we report every comparison over 5×5 evaluation runs.

Where strong classical baselines are included they often win: in the gate-based multivariate study of Hamhoum et al. [11] the QRC matches classical reservoir computing on Lorenz-63, but a nonlinear vector autoregression and a clustered ESN remain stronger in some settings.

3 Protocol

All runs use the `qrc_bench` pipeline (Section 7); every constant below is read from its source or from the stored records.

Task. Lorenz-96 with $K = 6$ variables on a ring, $\dot{x}_k = (x_{k+1} - x_{k-2})x_{k-1} - x_k + F$ with $F = 8$, is integrated by fourth-order Runge-Kutta with step 0.01; one sample is kept every 25 steps, after 1000 discarded samples, and each variable is standardised. A series has 2000 samples in the window cells and 3000 in the recurrent cells. The target is the next sample of all six variables. The score is the one-step NRMSE on the last 15 percent of the series, averaged over the six variables, each normalised by its test-slice standard deviation, so that NRMSE $= 1$ is the mean predictor. The readout is fit once on everything before the test slice, after a washout of 50 samples (window) or 400 (recurrent).

Quantum reservoir. The reservoir has $n_{\text{in}} = 6$ input qubits, one per variable, and $n_{\text{mem}} = 6$ memory qubits. Each input x is encoded as $R_Y(\text{clip}(s\,x, -\pi, \pi))|0\rangle$ with $s \in [0.1, \pi]$ tuned, and the system evolves for a tuned time $\tau \in [0.1, 10]$ under one of two all-to-all Hamiltonians,

$$H_{\text{XXZ}} = \sum_{i<j} (X_iX_j + Y_iY_j + \Delta\, Z_iZ_j) + \sum_i h_iZ_i, \quad h_i \sim \mathcal{U}(-W, W), \tag{1}$$

$$H_{\text{Ising}} = \sum_{i<j} J_{ij}\, X_iX_j + v \sum_i Z_i, \quad J_{ij} \sim \mathcal{U}(0, 1), \tag{2}$$

with $\Delta \in [-2, 2]$, $W \in [0.1, 10]$ and $v \in [0.1, 10]$ tuned and the disorder drawn from the reservoir seed. In the *window* cells the memory is reset: at every step the memory qubits start in $|0\rangle^{\otimes 6}$ and the last two inputs are driven in order, the input qubits being traced out between drives, so the QRC is a fixed feature map of a two-step window. In the *recurrent* cells the memory state is carried from step to step, and a feedback term shifts every input angle by $k_{\text{fb}} \tanh(\bar{m}_{t-1})$, where $\bar{m}_{t-1}$ is the mean measured $\langle Z \rangle$ of the previous step and $k_{\text{fb}} \in [0, 8]$ is tuned. The Z/ZZ readout gives $\langle Z_i \rangle$ and $\langle Z_iZ_j \rangle$ on all twelve qubits, 78 features; the +X/XX readout measures the same block a second time in the X basis, 156 features. The feedback signal and the memory state stay on the Z basis.

Shots. Every QRC feature in the eight cells is estimated from $1024\,q = 12288$ shots per step, split equally over the bases, so the second basis costs no extra shots. We use the restarting protocol [9]: each step's measurement is a fresh preparation of the same circuit, so the memory state carried forward is exact, while the features and the feedback signal are read from the sampled counts. The shot noise realisation is fixed by the reservoir seed.

Classical models. *Random features (RF)*: $\tanh(Az_t + b)$, where z_t stacks the same two-step window the QRC sees ($n = 12$ values), $A_{ij} \sim \mathcal{N}(0, \sigma^2/n)$, $b_i \sim \mathcal{U}(-\beta, \beta)$, $\sigma \in [10^{-3}, 10]$ and $\beta \in [0, 10]$ tuned, with the QRC's width. *Poly2* and *linear*: the complete degree-2 polynomial of the window (width 90) and the window itself (width 12), with only the ridge penalty tuned; both have no reservoir seed. *ESN*, recurrent cells only: a leaky echo-state network of the QRC's width, $r_t = (1 - a)\, r_{t-1} + a \tanh(W r_{t-1} + W_{\text{in}} x_t + b)$, with spectral radius in $[0.05, 1.5]$, leak $a \in [0.01, 1]$, input scale in $[0.01, 5]$, density in $[0.02, 1]$ and bias scale in $[0, 10]$ tuned. The window models remain as references in the recurrent cells.

Tuning. Every model is tuned by Optuna's TPE sampler [12] with the same sampler seed. The ridge penalty $\alpha \in [10^{-6}, 10^2]$ (log scale) is tuned for every model, on standardised features, and counts as a knob. The budget is matched by search density: a model with d knobs gets at least $25d$ trials and continues until 50 trials pass without a new best, with a cap of 300; every model in the eight cells stopped on the patience rule. The objective is the validation NRMSE over three rolling-origin folds (each 10 percent of the series, all before the test slice) and over three tuning data seeds × three tuning reservoir seeds. Each model's best setting is frozen and evaluated once on five fresh data seeds × five fresh reservoir seeds.

Statistics. For each QRC-versus-classical pair we form the paired difference on each of the 25 runs and on each of the 5 data-seed clusters. We count the runs and data seeds the QRC wins (differences within 5×10^{-4} are ties) and apply a paired t-test to the cluster means, Holm-corrected within the cell (three comparisons in a window cell, four in a recurrent cell). No correction is applied across the 28 tests of the grid; the random-feature verdict survives even a Bonferroni factor of 28 (largest corrected p: 3.2×10^{-3}), the two weakest Poly2 results (Holm $p = 0.011$ and 0.023) may not. For Poly2 and linear the five reservoir seeds of a data seed share one prediction, so only the cluster-level test is independent; we use it throughout.

4 Results

4.1 The eight cells

Table 1 and Fig. 1 give the test NRMSE of every model in every cell; the paired win counts and Holm p values are in Appendix A (Tables 7 and 8).

Table 1: Test NRMSE (mean ± s.d. over 25 runs, one step ahead, lower is better), QRC read from 12288 shots per step. Width is the QRC width, which the random features (RF) and the ESN match; "+X/XX" is Z/ZZ plus the X basis. Bold: best model in the cell. Poly2 and linear depend only on the data, so they repeat within a memory kind.

Kind	Res.	Readout	Width	QRC	RF	Poly2	Linear	ESN
window	XXZ	Z/ZZ	78	0.651 ± 0.018	0.402 ± 0.013	$\mathbf{0.373} \pm 0.013$	0.712 ± 0.010	–
window	XXZ	+X/XX	156	0.407 ± 0.018	$\mathbf{0.324} \pm 0.016$	0.373 ± 0.013	0.712 ± 0.010	–
window	Ising	Z/ZZ	78	0.572 ± 0.013	0.402 ± 0.013	$\mathbf{0.373} \pm 0.013$	0.712 ± 0.010	–
window	Ising	+X/XX	156	0.369 ± 0.012	$\mathbf{0.324} \pm 0.016$	0.373 ± 0.013	0.712 ± 0.010	–
recurrent	XXZ	Z/ZZ	78	0.523 ± 0.018	0.416 ± 0.016	0.392 ± 0.018	0.716 ± 0.010	$\mathbf{0.340} \pm 0.012$
recurrent	XXZ	+X/XX	156	0.421 ± 0.012	0.332 ± 0.017	0.392 ± 0.018	0.716 ± 0.010	$\mathbf{0.259} \pm 0.017$
recurrent	Ising	Z/ZZ	78	0.528 ± 0.012	0.416 ± 0.016	0.392 ± 0.018	0.716 ± 0.010	$\mathbf{0.340} \pm 0.012$
recurrent	Ising	+X/XX	156	0.414 ± 0.009	0.332 ± 0.017	0.392 ± 0.018	0.716 ± 0.010	$\mathbf{0.259} \pm 0.017$

Random features beat the QRC everywhere. In every cell the random-feature map has the lower test NRMSE on all 25 runs and all 5 data seeds, by 0.045 to 0.249, with Holm

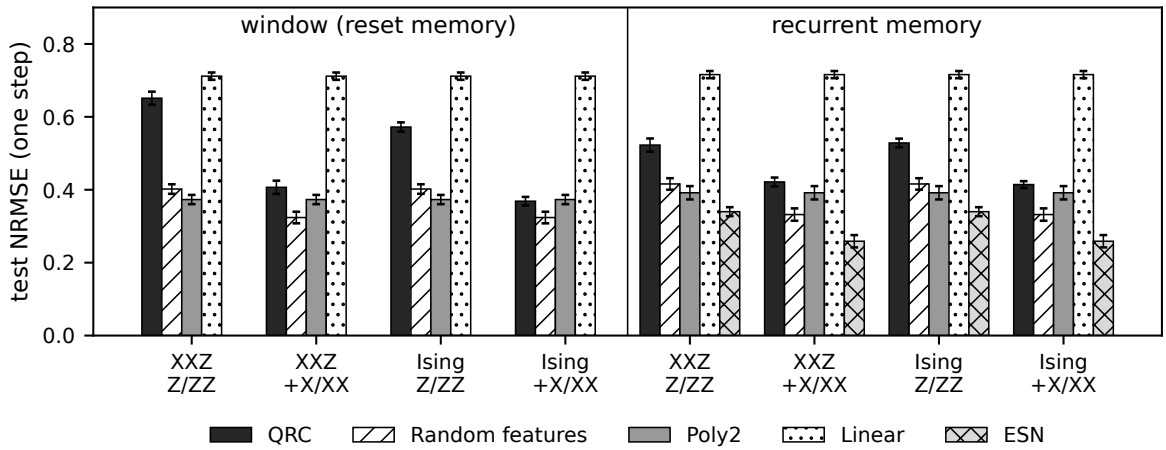


Figure 1: Test NRMSE per model in the eight cells (mean ± s.d. over 25 runs).

$p \leq 2.3 \times 10^{-4}$. It sees the same two-step window and has the same number of features, so neither input nor readout width explains the gap.

The second basis is the QRC's largest gain. Adding the X basis at half the shots per basis lowers the QRC's NRMSE in all four pairs (Table 2), and the largest of these drops is the largest change of the QRC between any two cells that differ in one factor. The width-matched classical models gain too, from the doubled width alone. In the window cells the QRC gains 2.6 to 3.1 times as much as the random features and the gap to them shrinks by 67 to 74 percent; in the recurrent cells the random features gain 74 to 83 percent as much as the QRC, so the gap shrinks only by 0.02 to 0.03. The QRC still loses on every seed.

Table 2: Effect of the second basis (same total shots). Columns 3 and 4: QRC test NRMSE; columns 5 to 7: change in mean test NRMSE from Z/ZZ to +X/XX for the QRC and the width-matched RF and ESN; columns 8 and 9: QRC minus RF.

		QRC NRMSE		change Z/ZZ → +X/XX			QRC − RF	
Kind	Res.	Z/ZZ	+X/XX	QRC	RF	ESN	Z/ZZ	+X/XX
window	XXZ	0.651	0.407	−0.244	−0.078	–	0.249	0.083
window	Ising	0.572	0.369	−0.203	−0.078	–	0.170	0.045
recurrent	XXZ	0.523	0.421	−0.101	−0.084	−0.081	0.107	0.089
recurrent	Ising	0.528	0.414	−0.114	−0.084	−0.081	0.112	0.082

The best QRC ties Poly2. In the best QRC cell (window, Ising, +X/XX) the QRC wins 14 of 25 runs and 4 of 5 data seeds against Poly2, but the cluster interval of the difference, $[-0.0090, 0.0001]$, contains zero and the Holm p is 0.157. In every other cell Poly2 has the lower mean. Every QRC beats the linear model on every run.

The ESN wins the recurrent kind. The ESN is ahead of the recurrent QRC on every run and data seed of the four recurrent cells, by 0.156 to 0.188, and at width 156 it is the best model in the grid. Its history is long only in name: as tuned, the spectral radius is about 0.05 and the leak 0.84 to 0.88, so the state keeps a weight of only 0.12 to 0.16 on its previous value plus a weak recurrent term. With two bases the recurrent QRC is no better than the reset one.

4.2 Shot noise

Table 3 and Fig. 2 compare the two XXZ, Z/ZZ cells with exact-expectation runs of the same protocol; in both conditions the QRC is tuned under the condition it is evaluated in, and the classical rows are identical. The shots cost the reset QRC 0.019 in NRMSE and the recurrent QRC 0.110. This is consistent with the feedback path: in the recurrent cells the feedback signal is read from the sampled counts, so measurement noise enters the drive and is carried in the memory state, while in the reset cells it touches only the readout. We did not run the ablation with sampled readout and exact feedback, so the mechanism is likely, not measured. The shots also change the verdict: with exact values the recurrent QRC ties the random features (it wins 17 of 25 runs and 4 of 5 data seeds, Holm $p = 0.434$); with shots it loses to them on every run.

Table 3: Exact expectation values against 12288 shots per step, XXZ reservoir, Z/ZZ readout: mean test NRMSE over 25 runs; "change" is shots minus exact.

	window, XXZ, Z/ZZ			recurrent, XXZ, Z/ZZ		
Model	exact	12288 shots	change	exact	12288 shots	change
QRC	0.632	0.651	+0.019	0.412	0.523	+0.110
Random features	0.402	0.402	0	0.416	0.416	0
Poly2	0.373	0.373	0	0.392	0.392	0
Linear	0.712	0.712	0	0.716	0.716	0
ESN	–	–	–	0.340	0.340	0

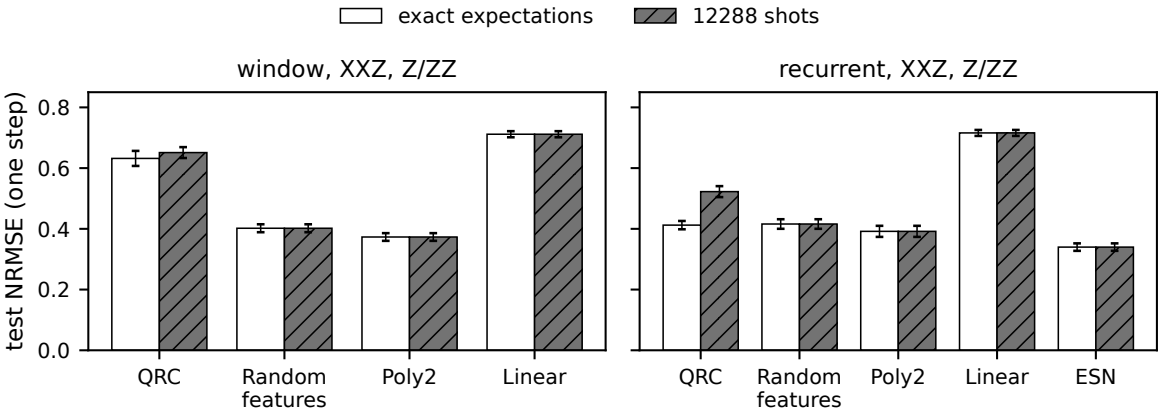


Figure 2: Exact expectation values (white) against 12288 shots per step (hatched), XXZ reservoir, Z/ZZ readout (mean ± s.d. over 25 runs). Only the QRC changes.

4.3 Is the search finished?

Local refinement. We repeated the best QRC cell with a refinement stage: after TPE stops, SPSA runs from each model's best TPE trial in the unit cube of its search space, with half the model's floor budget, rounded down to an even number, in further evaluations. The TPE stage reproduced the original run trial for trial. The refinement improved no model (Table 4, Fig. 3). Near the optimum the classical landscapes are flat (70 to 100 percent of TPE trials within 0.02 of the best validation NRMSE); the QRC's is rougher (37 percent), but a local search from its best point finds nothing either.

Widened ranges. In the two-basis Ising cells the tuned field sat at the top of its range ($v \leq 10$), and the ESN's spectral radius and bias scale at the bottom and top of theirs. We

Table 4: SPSA refinement after TPE, window, Ising, +X/XX. Knobs include the ridge penalty; "within 0.02": fraction of TPE trials within 0.02 of the best validation NRMSE.

Model	knobs	TPE trials	SPSA evals	best TPE val.	best SPSA val.	gain	within 0.02
QRC	4	197	50	0.3845	0.3852	0.0	37%
Random features	3	239	36	0.3570	0.3586	0.0	70%
Poly2	1	81	12	0.4069	0.4070	0.0	99%
Linear	1	111	12	0.7246	0.7246	0.0	100%

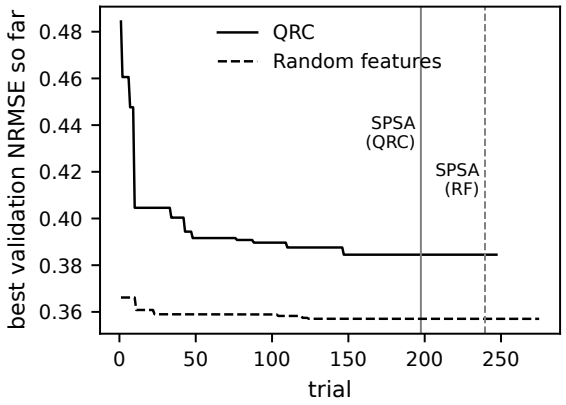


Figure 3: Best validation NRMSE so far, window, Ising, +X/XX, for the QRC (solid) and the random features (dashed). Vertical lines: where TPE stopped and SPSA started.

reran both cells with $v \in [0.1, 30]$, ESN spectral radius in $[0.01, 1.5]$, ESN bias scale in $[0, 30]$ and ridge penalty in $[10^{-9}, 10^{2}]$ for every model, all else unchanged (Table 5). The field moved well inside the new range and the ESN bias scale inside its range; the ESN spectral radius moved down but stays near its new lower bound. No verdict against the random features or the ESN changed: the QRC wins no data seed against either. Against Poly2 the window cell remains a tie, and the recurrent cell moves from a loss (Holm $p = 0.023$) to a tie (Holm $p = 0.335$). In the recurrent cell the random features stopped on the 300-trial cap rather than on the patience rule.

Table 5: Widened ranges, two-basis Ising cells: published ranges ("orig.") against widened ("wide"). Test NRMSE (mean over 25 runs), QRC data-seed wins (of 5) and Holm p against each model, and the tuned field v and ESN spectral radius and bias scale.

		test NRMSE				QRC seed wins, Holm p				ESN	
Kind	ranges	QRC	RF	Poly2	ESN	vs RF	vs Poly2	vs ESN	v	radius	bias
window	orig.	0.369	0.324	0.373	–	$0/5,\ 2.3 \times 10^{-4}$	4/5, 0.157	–	9.65	–	–
window	wide	0.369	0.321	0.373	–	$0/5,\ 1.7 \times 10^{-3}$	3/5, 0.421	–	16.51	–	–
recurrent	orig.	0.414	0.332	0.392	0.259	$0/5,\ 1.4 \times 10^{-4}$	0/5, 0.023	$0/5,\ 2.2 \times 10^{-5}$	9.66	0.052	9.80
recurrent	wide	0.399	0.331	0.392	0.251	$0/5,\ 7.4 \times 10^{-4}$	2/5, 0.335	$0/5,\ 1.9 \times 10^{-5}$	20.85	0.020	13.02

4.4 Free-running forecasts

We ran every frozen model autonomously from 20 starts in the test slice: the model predicts the next sample from the true history and then feeds each prediction back as its next input (recurrent models are first warmed on the true data; the QRC draws fresh shots every step). The valid prediction time (VPT) is the time before the error, the RMS over variables of the

error scaled by each variable's test-slice s.d., first exceeds 0.5, within 10 Lyapunov times. The largest Lyapunov exponent of the task's own integrator (Benettin's method on the same RK4 step) is $\lambda = 0.973$, so a Lyapunov time is 4.11 samples; the same routine gives 1.70 for $K = 40$, close to the value of about 1.68 reported for that system [13]. No model's mean VPT reaches one Lyapunov time (Table 6; the ESN is longest at 0.80), so these one-step readouts are poor autonomous forecasters at this sampling. The ordering against the stronger models holds: in every cell the random features outlast the QRC (significantly in 6 of 8; Appendix A), and in every recurrent cell the ESN outlasts both. With one basis the QRC falls to the level of the simpler models, tying the linear model in the window cells and Poly2 in the recurrent cells; with two it outlasts Poly2 on every run, because Poly2 diverges when fed its own output (mean NRMSE 8.3×10^4 after one Lyapunov time in the window cells). The coarse sampling also makes these horizons far shorter than those of Ahmed et al. [2], who sample every 0.01 time units and report VPTs of up to about 8.5 Lyapunov times on Lorenz-63; our VPTs compare the models of this grid with each other, not with that work.

Table 6: Free run: mean VPT in Lyapunov times ± s.d. over 25 runs, mean in samples in parentheses (20 starts per run, threshold 0.5). Bold: longest in the cell.

Res.	Readout	QRC	RF	Poly2	Linear	ESN
window						
XXZ	Z/ZZ	0.08 ± 0.03 (0.3)	$\mathbf{0.31} \pm 0.05$ (1.3)	0.24 ± 0.03 (1.0)	0.07 ± 0.04 (0.3)	–
XXZ	+X/XX	0.37 ± 0.06 (1.5)	$\mathbf{0.45} \pm 0.10$ (1.9)	0.24 ± 0.03 (1.0)	0.07 ± 0.04 (0.3)	–
Ising	Z/ZZ	0.13 ± 0.04 (0.5)	$\mathbf{0.31} \pm 0.05$ (1.3)	0.24 ± 0.03 (1.0)	0.07 ± 0.04 (0.3)	–
Ising	+X/XX	0.41 ± 0.09 (1.7)	$\mathbf{0.45} \pm 0.10$ (1.9)	0.24 ± 0.03 (1.0)	0.07 ± 0.04 (0.3)	–
recurrent						
XXZ	Z/ZZ	0.22 ± 0.05 (0.9)	0.29 ± 0.06 (1.2)	0.22 ± 0.02 (0.9)	0.08 ± 0.02 (0.3)	$\mathbf{0.62} \pm 0.10$ (2.5)
XXZ	+X/XX	0.34 ± 0.08 (1.4)	0.48 ± 0.07 (2.0)	0.22 ± 0.02 (0.9)	0.08 ± 0.02 (0.3)	$\mathbf{0.80} \pm 0.10$ (3.3)
Ising	Z/ZZ	0.22 ± 0.06 (0.9)	0.29 ± 0.06 (1.2)	0.22 ± 0.02 (0.9)	0.08 ± 0.02 (0.3)	$\mathbf{0.62} \pm 0.10$ (2.5)
Ising	+X/XX	0.38 ± 0.07 (1.6)	0.48 ± 0.07 (2.0)	0.22 ± 0.02 (0.9)	0.08 ± 0.02 (0.3)	$\mathbf{0.80} \pm 0.10$ (3.3)

5 Discussion

What would it take for the QRC to win here? Halving the shots per basis to add a basis helped in every cell, and most in the window cells, which suggests that there the limit is what is measured rather than how precisely. The gain is confounded with the doubled width, since we did not run a one-basis QRC of width 156, and the width-matched classical models gain from width too; a third basis must add more than width does. Wider ranges improved the QRC only within the verdicts it already had (Section 4.3). On hardware the restarting protocol is costly for the recurrent QRC: the circuit of step t replays all t drives, so one series of 3000 steps needs about 4.5×10^6 drives per shot round, each repeated 12288 times; the reset QRC needs two drives per step. The recurrent QRC is thus the more expensive scheme and, with two bases, not the better one. Weak-measurement and rewinding protocols [9] reduce the replay cost at the price of extra noise. Finally, the models are tuned for one step; a fair long-horizon test would sample finely and tune every model on the free-running metric.

Why earlier claims differ. The six patterns of Section 2 each predict a direction in our grid. *Resource units*: at equal feature width the random features win. *Baseline tuning*: in our first exact-expectation run of the recurrent XXZ cell the ESN had no bias term and reached 0.445 against the QRC's 0.412, the QRC winning 23 of 25 runs; adding the bias scale to the ESN's search, with the QRC's result unchanged, gives 0.340 and an ESN win on every run. *QRC against QRC*: the second basis is a real gain over the one-basis QRC, which would read

as a success against that baseline alone. *Capacity metrics*: we report forecast error only. *Exact expectations*: Section 4.2 shows a tie under exact values becoming a loss on every seed under shots. *Few seeds*: the QRC's s.d. over runs is 0.009 to 0.018, so single-run gaps smaller than that are not interpretable; the margins we report are larger and hold on every data seed.

6 Limitations

The study has one task (Lorenz-96, $K = 6$, $F = 8$), one system size ($q = 12$), one input window (two steps), two reservoir families with all-to-all couplings, one shot budget, and one tuning horizon (one step). The free-running test uses coarse sampling, so its VPTs resolve only a few steps, and a single error threshold. The shot-noise contrast rests on one single-basis XXZ pair per memory scheme, the feedback-noise mechanism is not tested by an ablation, and the refinement stage and the widened ranges were each tested in one or two cells. The Holm correction is applied within cells, not across the grid. Shots are sampled from the exact state, with no gate, readout or decoherence noise, so hardware would be harder on the QRC, not easier. For scale, the no-change forecast scores NRMSE 1.13 to 1.16 on the test slices: at this sampling the last value is a worse forecast than the mean. The results say nothing about larger reservoirs, other encodings, quantum input data or other tasks.

7 Reproducibility

All runs use the `qrc_bench` pipeline (https://github.com/pandey-tushar/Quantum_Chaos_solver). Each record stores its protocol, the trial history of every model, the per-run test scores, package versions, the GPU and the pipeline's git commit (Table 11, Appendix A). The runs used one NVIDIA GeForce RTX 2050 GPU with 4 GB of memory; the comparison records take 22.8 hours of wall time in total, almost all of it in the QRC's tuning, and the eight free runs 0.66 h. Two records ran from changes later committed: the refinement record carries commit `bccf36d`, its SPSA stage committed as `df03eeb`; the free-run records carry `df03eeb`, the free-run code committed as `752b50b`. The tables and inline numbers of this paper are generated by `make_tables.py` and its figures by `make_figures.py`, both reading the records directly.

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A Paired statistics and records

Table 7: One step: QRC wins against each classical model, runs (of 25) and, in parentheses, data seeds (of 5). †: the classical model has no reservoir seed, so run-level counts are not independent.

Kind	Res.	Readout	vs Random features	vs Poly2	vs Linear	vs ESN
window	XXZ	Z/ZZ	0/25 (0/5)	0/25 (0/5)†	25/25 (5/5)†	–
window	XXZ	+X/XX	0/25 (0/5)	0/25 (0/5)†	25/25 (5/5)†	–
window	Ising	Z/ZZ	0/25 (0/5)	0/25 (0/5)†	25/25 (5/5)†	–
window	Ising	+X/XX	0/25 (0/5)	14/25 (4/5)†	25/25 (5/5)†	–
recurrent	XXZ	Z/ZZ	0/25 (0/5)	0/25 (0/5)†	25/25 (5/5)†	0/25 (0/5)
recurrent	XXZ	+X/XX	0/25 (0/5)	0/25 (0/5)†	25/25 (5/5)†	0/25 (0/5)
recurrent	Ising	Z/ZZ	0/25 (0/5)	0/25 (0/5)†	25/25 (5/5)†	0/25 (0/5)
recurrent	Ising	+X/XX	0/25 (0/5)	1/25 (0/5)†	25/25 (5/5)†	0/25 (0/5)

Table 8: One step: Holm-corrected p of the QRC against each classical model (paired t-test on the five data-seed means, Holm within the cell).

Kind	Res.	Readout	vs Random features	vs Poly2	vs Linear	vs ESN
window	XXZ	Z/ZZ	2.7×10^{-6}	2.7×10^{-6}	1.3×10^{-4}	–
window	XXZ	+X/XX	1.3×10^{-5}	6.7×10^{-5}	1.1×10^{-7}	–
window	Ising	Z/ZZ	1.3×10^{-5}	1.3×10^{-5}	3.1×10^{-5}	–
window	Ising	+X/XX	2.3×10^{-4}	0.157	6.5×10^{-8}	–
recurrent	XXZ	Z/ZZ	7.4×10^{-5}	7.4×10^{-5}	1.2×10^{-6}	1.9×10^{-6}
recurrent	XXZ	+X/XX	1.1×10^{-4}	0.011	2.3×10^{-7}	2.2×10^{-5}
recurrent	Ising	Z/ZZ	5.1×10^{-5}	5.1×10^{-5}	2.3×10^{-7}	2.3×10^{-6}
recurrent	Ising	+X/XX	1.4×10^{-4}	0.023	1.2×10^{-7}	2.2×10^{-5}

Table 9: Free run: QRC wins on VPT against each classical model, as in Table 7.

Kind	Res.	Readout	vs Random features	vs Poly2	vs Linear	vs ESN
window	XXZ	Z/ZZ	0/25 (0/5)	0/25 (0/5)†	12/25 (2/5)†	–
window	XXZ	+X/XX	1/25 (0/5)	25/25 (5/5)†	25/25 (5/5)†	–
window	Ising	Z/ZZ	0/25 (0/5)	0/25 (0/5)†	19/25 (5/5)†	–
window	Ising	+X/XX	8/25 (1/5)	25/25 (5/5)†	25/25 (5/5)†	–
recurrent	XXZ	Z/ZZ	2/25 (0/5)	10/25 (1/5)†	25/25 (5/5)†	0/25 (0/5)
recurrent	XXZ	+X/XX	2/25 (0/5)	25/25 (5/5)†	25/25 (5/5)†	0/25 (0/5)
recurrent	Ising	Z/ZZ	4/25 (1/5)	12/25 (3/5)†	25/25 (5/5)†	0/25 (0/5)
recurrent	Ising	+X/XX	2/25 (0/5)	25/25 (5/5)†	25/25 (5/5)†	0/25 (0/5)

Table 10: Free run: Holm-corrected p on VPT, as in Table 8.

Kind	Res.	Readout	vs Random features	vs Poly2	vs Linear	vs ESN
window	XXZ	Z/ZZ	3.5×10^{-4}	3.5×10^{-4}	0.597	–
window	XXZ	+X/XX	0.020	1.0×10^{-3}	1.0×10^{-3}	–
window	Ising	Z/ZZ	2.8×10^{-4}	8.5×10^{-4}	0.076	–
window	Ising	+X/XX	0.035	1.9×10^{-3}	4.1×10^{-4}	–
recurrent	XXZ	Z/ZZ	0.082	0.937	9.6×10^{-4}	4.8×10^{-4}
recurrent	XXZ	+X/XX	9.5×10^{-3}	9.5×10^{-3}	3.5×10^{-3}	4.2×10^{-4}
recurrent	Ising	Z/ZZ	0.153	0.796	1.6×10^{-3}	1.6×10^{-3}
recurrent	Ising	+X/XX	6.8×10^{-3}	3.6×10^{-3}	1.4×10^{-3}	9.1×10^{-4}

Table 11: Records, pipeline commit (`environment.git_commit`), total wall time (`total_seconds`) and the QRC's tuning time. `l96_recurrent_xxz` is the earlier exact run whose ESN had no bias knob.

Record	commit	total (h)	QRC tuning (h)
`l96_window_xxz_shots`	`a902b04`	0.39	0.35
`l96_window_xxz_zx_shots`	`bccf36d`	0.51	0.43
`l96_window_ising_shots`	`a902b04`	0.64	0.59
`l96_window_ising_zx_shots`	`bccf36d`	1.11	1.03
`l96_recurrent_xxz_shots`	`a902b04`	1.64	1.59
`l96_recurrent_xxz_zx_shots`	`bc6e881`	2.92	2.79
`l96_recurrent_ising_shots`	`bc6e881`	2.14	2.09
`l96_recurrent_ising_zx_shots`	`bccf36d`	2.72	2.57
`l96_window_xxz_exact`	`752b50b`	0.28	0.26
`l96_recurrent_xxz_exact`	`752b50b`	1.70	1.65
`l96_recurrent_xxz`	`4538057`	3.51	3.31
`l96_window_ising_zx_refine`	`bccf36d`	1.39	1.29
`l96_window_ising_zx_wide`	`752b50b`	0.88	0.76
`l96_recurrent_ising_zx_wide`	`752b50b`	2.94	2.78